

Fire Detection and Water Spraying Drone Using ESP32

Ikram Hussen
Department of CSE
Leading University
sylhet, Bangladesh
ikram@heptas.co.uk

Baishakhi Deb Momo
Department of CSE
Leading University
Sylhet, Bangladesh
baishakhideb43@gmail.com

Nishat Tanjum Ahmed
Department of CSE
Leading University
Sylhet, Bangladesh
pakhinishu5@gmail.com

Shomita Baishya
Department of CSE
Leading University
Sylhet, Bangladesh
hridibaishya@gmail.com

Abdulla Hal Basid
Department of CSE
Leading University
Sylhet, Bangladesh
abdullahalbasid864@gmail.com

Muhin Ahmed
Department of CSE
Leading University
Sylhet, Bangladesh
wasiqarmaan00@gmail.com

Abstract—Fire accidents cause serious damage to life and property, especially in areas where immediate human intervention is risky. To address this problem, this project proposes a microcontroller-based fire detection and water spraying system using a drone-inspired prototype. The system uses multiple flame sensors to detect the presence and direction of fire over a wide angular range. An ESP32 microcontroller processes the sensor data and communicates with an Arduino-based control unit. Two servo motors provide horizontal and vertical movement of a water nozzle, allowing accurate targeting of the fire source. A laser module is used for aiming assistance, while a ground-based water pump supplies water through a pipe to the nozzle. The system can be controlled and monitored through a web interface, enabling remote operation. Experimental results show that the system detects fire within 2–3 seconds over a 270° coverage range and successfully suppresses fire in controlled indoor conditions. This prototype demonstrates a low-cost, scalable, and safer approach to early fire detection and suppression, reducing the need for direct human involvement in hazardous environments.

Index Terms—Fire detection, Flame sensor, ESP32, Servo-controlled nozzle, Remote water spraying

I. INTRODUCTION

Fire incidents pose a serious threat to human life, property, and the environment worldwide, especially in industrial, residential, and remote areas where rapid human intervention is often difficult or unsafe. Delayed detection and response can lead to significant loss and endanger personnel, highlighting the need for automated fire detection and suppression systems. Traditional firefighting methods rely on manual observation and intervention, which are not only slow but also expose firefighters to hazardous conditions. In addition, in densely populated or remote regions, immediate access to fire sites may be limited, further increasing the risk of extensive damage.

The specific problem addressed by this research is the lack of a low-cost, integrated system capable of detecting fire in real time and suppressing it using a controlled water spray mechanism. Existing systems are often expensive, fixed in location, or focused solely on detection without providing

immediate suppression capabilities. Many rely on proprietary hardware, complex network configurations, or manual control, making them unsuitable for low-resource settings or small-scale applications.

Recent advancements in microcontrollers, sensors, and wireless communication technologies, particularly the ESP32, enable the development of compact, cost-effective, and remotely controllable systems. Integrating flame sensors with servo-controlled nozzles and a water pumping mechanism allows for precise directional response, while web-based interfaces enable remote monitoring and control. Using multiple flame sensors covering a wide angular range improves detection accuracy, and servo motors provide both horizontal and vertical movement of the nozzle to ensure effective water targeting.

The proposed system combines flame detection, actuator control, and remote operation in a single platform, forming a prototype for a drone-based fire detection and suppression system. The water pump supplies water from a ground reservoir to the nozzle, while the laser module enhances aiming accuracy. By integrating real-time sensing, actuation, and web-based control, the system reduces human exposure to dangerous environments, offers scalable deployment, and provides a low-cost alternative to conventional firefighting solutions.

The main contributions of this research include the design and implementation of a **microcontroller-based, servo-controlled water spraying system**, the development of a web interface for remote operation, and the demonstration of a prototype capable of detecting and suppressing fire in a controlled environment. This work provides a foundation for future development of automated, IoT-enabled firefighting systems, which can be deployed in industrial, residential, and hazardous environments to improve safety, reduce response times, and minimize property damage.

II. LITERATURE REVIEW

Fire detection and suppression systems have been extensively studied due to their critical role in protecting life, property, and the environment. Traditional systems, such as smoke detectors, heat sensors, and fixed sprinklers, provide basic protection but often face limitations in response time, directional targeting, and automation. These limitations can result in delayed fire suppression and increased risk to both people and property.

Recent developments in microcontroller-based systems and IoT technologies have enabled the creation of intelligent and responsive fire detection solutions. Flame sensors and temperature sensors integrated with microcontrollers allow real-time monitoring and automated decision-making. Servo-controlled mechanisms have been employed to accurately direct water jets toward fire sources, improving the effectiveness of suppression systems.

The integration of multiple sensors enhances detection accuracy and reduces false alarms. Combining real-time sensor data with actuator control allows precise targeting of fire, ensuring efficient water usage and faster suppression. Additionally, incorporating web-based or remote monitoring systems provides flexibility, allowing operators to control and supervise the system from safe distances.

Despite these advancements, many existing solutions remain expensive, complex, or limited to fixed installations. Portable or small-scale systems capable of both detecting fire and actively suppressing it are relatively scarce. There is also a need for systems that can provide comprehensive coverage over a wide angular range and allow accurate directional spraying without extensive human intervention.

The proposed system builds on these developments by integrating multiple flame sensors, ESP32 and Arduino microcontrollers, servo-controlled nozzle movement, and a water pump into a compact, low-cost, and remotely controllable prototype. This approach addresses the limitations of previous systems by providing both fire detection and active suppression in a practical, scalable, and user-friendly design. The system demonstrates a safer and more efficient method for early fire detection and suppression, suitable for industrial, residential, and remote applications.

III. METHODOLOGY

The proposed system is a microcontroller-based fire detection and water spraying prototype designed to detect fire, aim a nozzle accurately, and suppress fire using water. The system integrates flame sensors, ESP32 and Arduino microcontrollers, servo motors, a laser module, a water pump, and a web-based interface for remote monitoring and control. The primary objective is to provide a low-cost, automated, and scalable solution for early fire detection and suppression.

A. Overview

The system is divided into four main components: sensing, control, actuation, and remote monitoring. The sensing unit consists of two 5-way flame sensors covering approximately

270 degrees to detect the presence and direction of fire. The control unit comprises an ESP32 microcontroller and an Arduino UNO, where the ESP32 processes sensor signals and calculates fire location, and the Arduino controls the servo motors and water pump. The actuation unit includes horizontal and vertical servo motors, a water pump, a nozzle, and a laser module that together perform the physical operation of directing water toward the fire. Finally, the remote monitoring unit allows operators to supervise and control the system via a web-based interface, enhancing safety and operational flexibility.

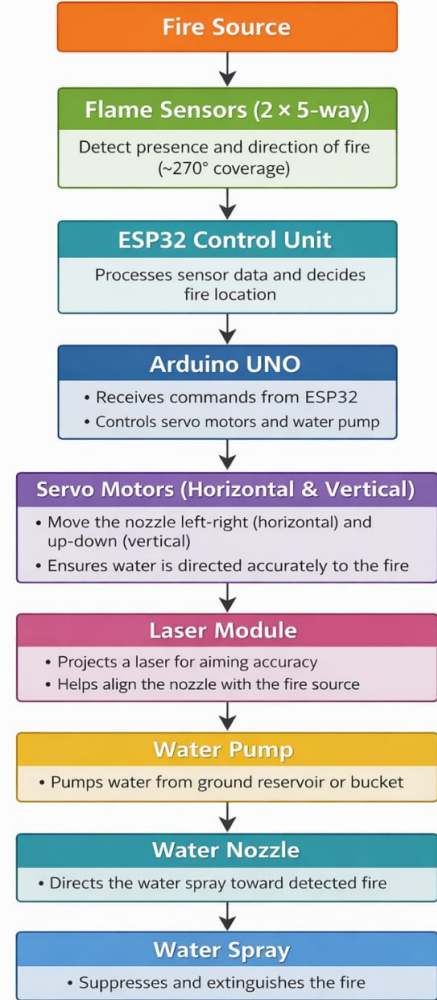


Fig. 1. Functional Block Diagram

B. Hardware Components

The hardware components of the system include the flame sensors, ESP32 and Arduino UNO microcontrollers, servo motors, laser module, water pump, nozzle, power supply, and supporting components such as breadboards and jumper wires. The flame sensors detect fire and provide analog or digital signals to the ESP32. The ESP32 processes this data and

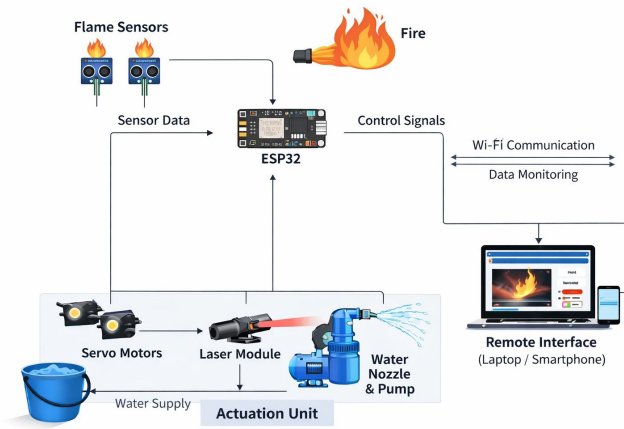


Fig. 2. System Architecture

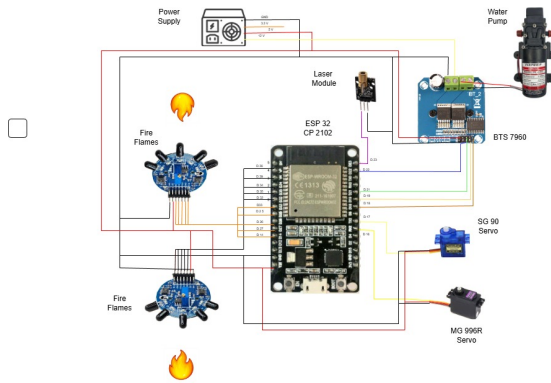


Fig. 3. The schematic design of Fire Detection and Water Spraying Drone

communicates with the Arduino to actuate the servo motors and pump. The horizontal and vertical servo motors move the nozzle precisely toward the fire, while the laser module aids in accurate targeting. The water pump delivers water from a reservoir through the nozzle to suppress the fire. A dedicated power supply ensures stable operation of all components, and the modular hardware design allows for easy replacement or addition of components.

C. Design of Wireless Network

The wireless network is implemented using the ESP32 microcontroller's Wi-Fi capabilities, enabling communication between the drone system and the web-based interface. The ESP32 sends real-time sensor data, including fire detection status and nozzle position, to the web server. Control commands from the operator, such as adjusting nozzle direction or activating the water pump, are transmitted back to the ESP32. This design ensures that the system can operate remotely without physical interaction, providing safety for the operator and flexibility in controlling the system from a distance. The network is designed to be reliable and low-latency to allow near real-time operation.

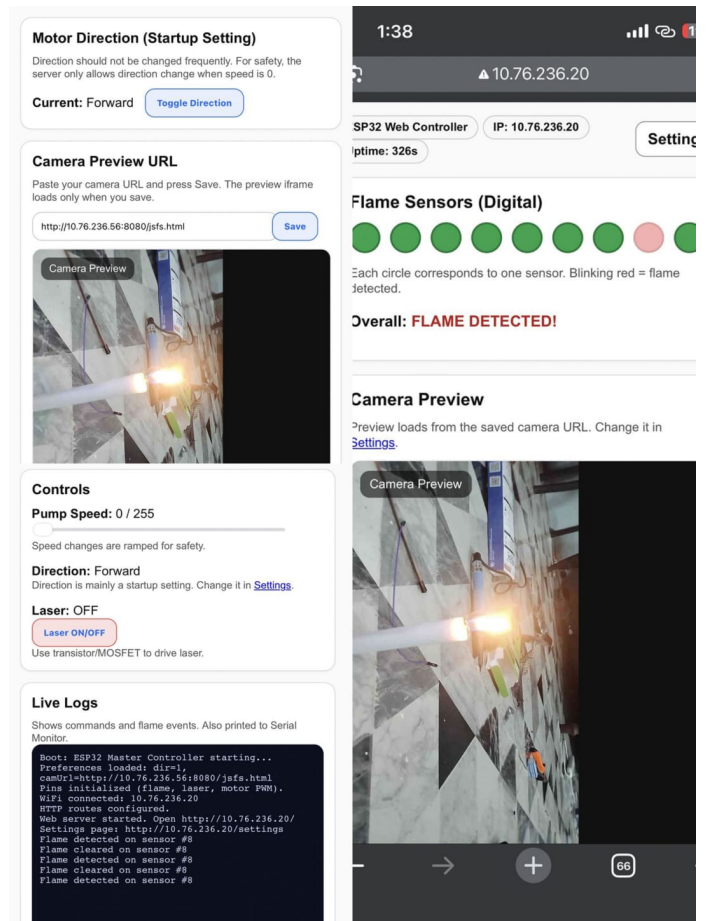


Fig. 4. Snapshot of the webpage

D. Website Design for Remote Control

The web interface is designed to allow remote monitoring and manual control of the system. The website displays real-time sensor data, including fire detection alerts and nozzle orientation, and provides controls for adjusting the horizontal and vertical movement of the nozzle and activating the water pump. The interface is user-friendly, providing intuitive buttons and sliders to operate the system safely. The web-based design allows the operator to monitor the fire situation and control the drone from any location with internet access. Security measures such as authentication and HTTPS can be incorporated to protect the system from unauthorized access.

IV. RESULTS AND DISCUSSION

A. Results

The microprocessor-based fire detection and water spraying drone prototype was designed and implemented using low-cost embedded system components. The system integrates flame sensors, an ESP32 microcontroller, servo motors, and a water pumping mechanism to detect fire and direct water toward the detected source.

Two 5-way flame sensors were mounted on the upper frame of the prototype and interfaced with the ESP32 to continuously monitor the surrounding area. While full 360° fire detection typically requires four sensors, the use of two sensors provided approximately 270° coverage, which was sufficient for prototype validation. Upon flame detection, the ESP32 processed sensor data to estimate the fire direction.

Two MG995 continuous rotation servo motors were used to control nozzle movement, with one servo enabling horizontal rotation and the other providing vertical motion. This configuration allowed accurate alignment of the nozzle toward the fire source. A laser diode module was added near the nozzle to improve visual targeting accuracy.

Water was supplied through a pipe connected to a ground-based pump and reservoir. The system also incorporated a web-based control interface using the ESP32's wireless capability, allowing remote manual control of nozzle direction and pump activation. Experimental results confirm effective fire detection, directional response, and reliable water spraying.

B. Discussion

The prototype demonstrates the feasibility of embedded systems for fire detection and suppression. The ESP32's built-in Wi-Fi enabled real-time monitoring and manual control via a web interface. Limitations include using only two flame sensors, which reduced coverage and precision, and reliance on a ground-based pump, limiting mobility. High power consumption affected servo performance, and continuous rotation servos without positional feedback reduced angular accuracy. Future improvements include four sensors, onboard water supply, power regulation, and positional servos or encoders. Despite these, the system validates effective fire detection and response.

V. FUTURE WORK AND CONCLUSION

A. Future Work

Future developments of the fire detection and water spraying drone aim to integrate machine learning algorithms for intelligent fire detection. Leveraging the existing onboard camera, the system will be able to automatically identify fire locations without relying solely on flame sensors. This enhancement will enable autonomous operation alongside manual control via the web-based interface, improving response speed, accuracy, and overall system efficiency. Additionally, incorporating AI-based fire detection can reduce false alarms, optimize water usage, and enable the drone to prioritize critical fire zones during emergency situations.

B. Conclusion

In this project, a microcontroller-based fire detection and extinguishing system using a drone mechanism was successfully designed and demonstrated. The system integrates 5-way flame sensors, ESP32, and Arduino UNO to detect fire, determine its direction, and suppress it using a water spraying mechanism.

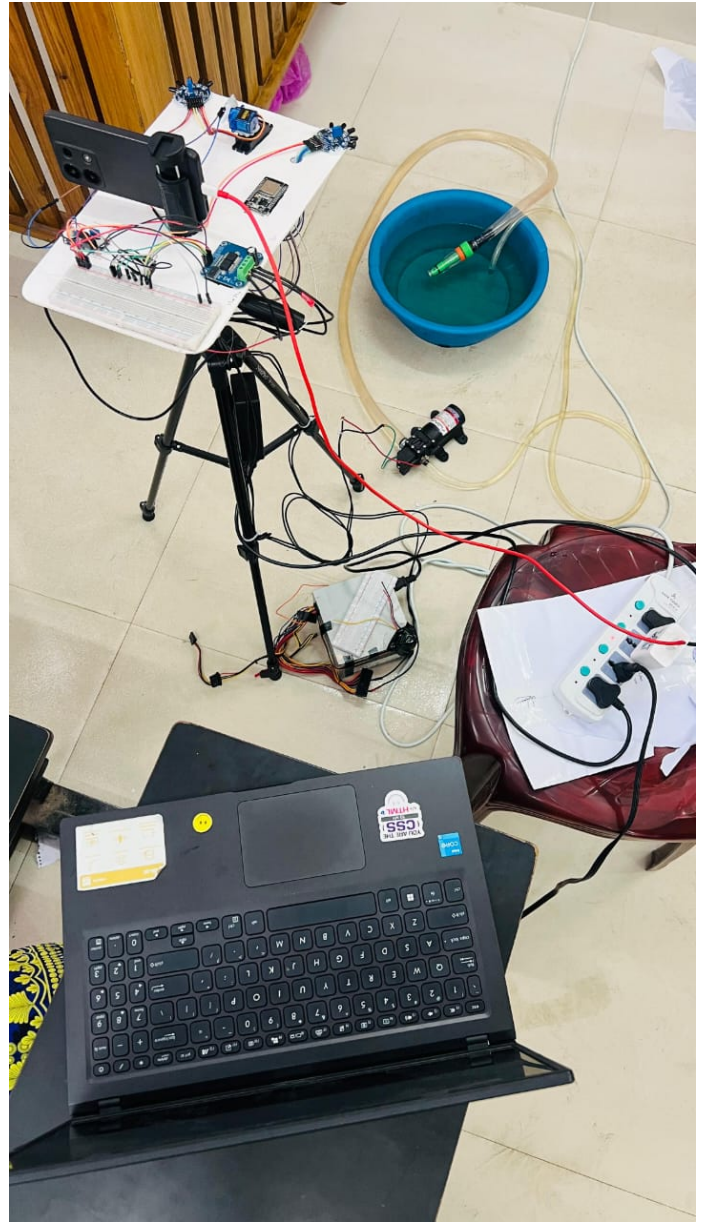


Fig. 5. The hardware implementation

The ESP32 is responsible for real-time fire detection and wireless communication, enabling remote monitoring and control through a web interface. The Arduino UNO controls the pan and tilt servo motors, laser module, and water pump, ensuring accurate targeting and efficient fire suppression. The laser module improves precision by indicating the exact spray direction, while the dual-servo setup allows wide-angle coverage.

The prototype proves that early fire detection and controlled extinguishing can be achieved with low-cost, easily available components. This system can reduce human risk in hazardous fire situations and demonstrates a practical application of embedded systems, IoT, and automation in fire safety.

Overall, the project meets its objectives and validates the feasibility of a drone-assisted fire-fighting system for small-scale and indoor environments.

REFERENCES